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Addis Ababa University

College of health Science, School of medicine

Module -I Assessment - II for year I medicine students

I. Choose the best answer among the given alternatives. Time: 1:30 Hr

1. Which is true about the negative feedback control system?

- A. The change between the set point and deviated value made greater
- B. The changing process and the response has opposite direction
- C. The responses are made through anticipation
- D. The response from the body is established before the value is changed
- E. Less involved in the normal body processes

2. Which of the following is not the ability of the homeostasis control system of the body?

- A. Detecting the change
- B. Responding for the change
- C. Comparing the normal and the changed value
- D. Sending the information into integrator
- E. None of the above

3. Which of the following is not an example of positive feedback homeostasis control system?

- | | |
|-----------------------------|---|
| A. End product facilitation | D. Luteinizing hormone surge |
| B. Cell depolarization | E. Oxytocine concentration during labor |
| C. Temperature regulation | |

4. Our immune cells do not attack the self cells in the normal condition. This is because of:

- A. Identity membrane proteins on our cells
- B. Identity membrane carbohydrates on our cells
- C. Receptor membrane proteins on our cells
- D. Transporter membrane proteins on our cells
- E. Identity membrane integrine proteins on our cells

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6. Which of the following is the homeostatic value of human body?

- A. Blood PH of 9.5
- B. Core body temperature of 41°
- C. Blood sodium of 135-145mEq/L
- D. Blood calcium level of 1-4mg/dl of blood
- E. Human heart rate of 105 beats /minute

7. On which of the following level changes is made to cope up the challenges we exposed except one during allostatic response

- A. Genetic level
- B. Nervous system level
- C. Endocrine system level
- D. Cellular level
- E. None of the above

8. Which of the following is the true concept of allostasis?

- A. Set point variation is an error signal
- B. Constancy doesn't mean normal
- C. Changed value necessary indicates disease
- D. Counter acting is basic principle in allostasis
- E. Steady at the same value

9. At what step blood is said to be clotted

- A. When platelets are aggregated in the ruptured area of blood vessels
- B. When fibrin forms a cross bridge among aggregated platelets
- C. When platelet adhere with the endothelial cells
- D. When blood vessels are ruptured by tissue damage
- E. When blood vessels are constricted by sympathetic action

10. Which of the following is an example of homeostasis?

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11. What changes you don't expect in a man elevated to higher altitude in the first 4-5 days
- A. Increased respiratory rate
 - B. Pulmonary artery resistance reduction
 - C. Increased red blood cells count
 - D. Widen chest and strengthen intercostals muscles
 - E. Peripheral chemoreceptor activation
12. Which of the following changes not made in the body of human being during allostatic responses
- A. Immune suppression
 - B. Body coordination alteration
 - C. Structural changes
 - D. Functional changes
 - E. None of the above
13. Which of the following is true about the human body adjustment
- A. Prolonged allostatic response not contribute for organ damage
 - B. Fulfillment of the new demand by organism helps the organism for adaptation
 - C. Body functional changes for the new demand is always positive
 - D. Maintaining the goodwill is through constancy in allostatic concept
 - E. Hippocampal neuronal plasticity has no advantage for organism adaptation
14. Which of the following is the allostatic response but not allostatic load for the new demand
- A. Enlargement of thyroid gland in cold environment
 - B. Stress induced reduction of memory cells number in the brain
 - C. Glucose induced insulin production from pancreatic gland
 - D. Cessation of farther release of chemicals from the nerve terminal
 - E. Bleeding induced increase blood pressure and heart rate
15. Which of the following is true about the blood pressure regulation?
- A. Stress relaxation increases blood pressure
 - B. Baroreflex responses regulate blood pressure at intermediate time
 - C. The sympathetic outflow is reduced to decrease the blood pressure
 - D. Parasympathetic nerve effects on heart is bigger to increase the blood pressure

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16. Select the correct concept of closed and open loop control system
- A. The response from the body has effects on integrator center in open loop control system
 - B. Temperature regulation is an example of open loop control system
 - C. Closed loop is feedback control systems.
 - D. Closed loop system is more applicable in the biological process
 - E. Output has negative effects on the inputs that bring changes on the body values in open loop control system
17. Which of the following is true?
- A. Physiology studies about the structure of the body
 - B. Physiology studies the purpose of process at systemic level only
 - C. The scope of physiology is starting from microscopic structure to the organism level
 - D. The mechanistic approach of physiology studies the purpose of process in the body
 - E. The concepts in physiology do not contribute for the current medicine
18. Which of the following is not homeostatically regulated variable?
- A. Temperature of the body
 - B. Blood glucose level
 - C. Blood osmolarity
 - D. The blood pH
 - E. None of the above
19. Assume your glucose level is elevated to 160mg/dl of blood 20 minutes after carbohydrate meal. However, after 2 hours the glucose level is return back nearly into the normal. Then select the true idea given bellow
- A. Glucagon is produced following this carbohydrate meal
 - B. Alpha cells of the pancreatic gland are activated following this meal
 - C. The sensor for high blood glucose level following this carbohydrate meal is in the brain
 - D. The effectors where appropriate response made for this change are muscles cells
 - E. Glucose is released into the blood from tissues following this carbohydrate meal

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20. Assume you eat more fat meal frequently and the leptin production from the white adipocytes is increased. The leptin has receptors in the hypothalamus of the brain so food intake is going to be reduced and energy expenditure is elevated. Which of the following is true?

- A. This is an example of the feedforward control loop system
- B. Responses have no effects on the inputs
- C. This is an example of positive feedback control system
- D. This is a kind of anticipation responses made in the body
- E. The stimulus that bring the change is resisted

21. Which of the following is true?

- A. Most of the processes in the body are controlled voluntarily
- B. Cells in one tissue type should have unrelated structure and function
- C. All the systems are integrated to fulfill the demands of human body
- D. The set points of human body should be always constant to maintain organism's function
- E. Most of the biological processes in human body are controlled by open loop system

22. Which of the following is the negative feedback homeostasis response?

- A. Luteinizing hormone surge
- B. Oxytocine production during cervix relaxation
- C. Platelet aggregation on the ruptured blood vessels
- D. Cold sensitive neurons activation by T° reduction
- E. Cardiac contraction strength following 3L bleeding

23. Which of the following is the immediate homeostatic response for the change made in body?

- A. Cold induced enlargement of thyroid gland
- B. Hot temperature induced increased sweat gland effectiveness
- C. Gene expression alteration for a different behavior to be observed
- D. Cold exposure induced surface vessels constriction
- E. Environmental change induced acclimatization

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25. One of the following is incorrect regarding physiological buffers:
- A. Phosphate buffer contributes to intracellular body buffering system
 - B. Bicarbonate buffering system is more efficient than phosphate buffering
 - C. Protein buffer system contributes to both intracellular and extracellular buffering of body fluids
 - D. Excessive production of carbon dioxide is never be the cause of acidosis in the system
 - E. Bicarbonate buffering system is mostly extracellular
26. Chala is a first year medical student, he is nervous because the modular exam is difficult for him and he is breathing rapidly (hyperventilation). What do you expect out of the following?
- A. Metabolic acidosis, compensated by both renal and respiratory system
 - B. Metabolic alkalosis, compensated with renal as well as respiratory
 - C. Respiratory acidosis, compensated with both respiratory and renal system
 - D. Respiratory alkalosis, compensated by renal system
 - E. Both metabolic and respiratory acidosis, compensated by renal and respiratory systems
27. The Henderson-Hassel Balch equation is represented as:
- A. $\text{pH} = \text{pK}_a + \log(\text{A}^-/\text{HA})$
 - B. $\text{pH} = \text{pK}_a + \log(\text{HA}/\text{A}^-)$
 - C. $\text{pH} = \text{pK}_a - \log(\text{A}^-/\text{HA})$
 - D. $\text{pK}_a = \text{pH} - \log(\text{HA}/\text{A}^-)$
 - E. $\text{pH} = \text{pK}_a + \log(\text{H}^+/\text{HA})$
28. Which of the following is the most important chemical buffer of the plasma?
- A. $\text{HCO}_3^-/\text{H}_2\text{CO}_3$
 - B. $\text{HPO}_4^{2-}/\text{HPO}_4^-$
 - C. Organic phosphate Esters
 - D. Proteins
 - E. Hemoglobin
29. Which of the following the laboratory results indicates compensated metabolic alkalosis?

30. Which one of the following is incorrect about the plasma membrane?
- A. Cell membrane is bio asymmetric in nature
 - B. Plasma membrane is a fluid mosaic
 - C. Flip-flop movement occur always occur
 - D. Membranes behave like viscous at ordinary temperatures
 - E. Bio membranes are fragile
31. Which of the following phospholipids are localized mostly in the outer leaflet of the membrane lipid bilayer?
- A. Choline phosphoglycerides
 - B. Inositol phosphoglycerides
 - C. Ethanolamine phosphoglycerides
 - D. Serine phosphoglycerides
 - E. Phosphatidic acid
32. Which of the following is not a function of membrane proteins?
- A. Provide cellular identification tags
 - B. Attach the membrane to the cytoskeleton
 - C. Serve as enzymes
 - D. Cell-cell communications
 - E. Transports lipophilic molecules across the membrane
33. Which of the following inhibits ions from passing through a lipid bilayer of cell membrane?
- A. The watery environment on either side of the lipid bilayer
 - B. The hydrophilic exterior of the lipid bilayer
 - C. The hydrophobic interior of the lipid bilayer
 - D. The lipophobic exterior of lipid bilayer
 - E. All of the above
34. Membrane phospholipids:
- A. Have hydrophilic heads that face outward and are exposed to water

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35. Most important reason for the special properties of water is:

- A. The covalent bonding pattern in water molecule
- B. The bond angle between the two hydrogen atoms in water
- C. Hydrogen bonding between water molecules
- D. Water can be immediately ionized at room temperature
- E. Water can exist as solid

36. The branch points of amylopectin in starch granules are connected by:

- A. α (1 $\rightarrow$ 4)
- B. β (1 $\rightarrow$ 4)
- C. α (1 $\rightarrow$ 6)
- D. β (1 $\rightarrow$ 6)
- E. α (1 $\rightarrow$ 2)

37. Which of the following is an example for heteropolysaccharide?

- A. Starch
- B. Glycogen
- C. Cellulose
- D. Chitin
- E. Hyaluronic acid

38. Which of the following is not an organic biomolecule?

- A. Carbohydrates
- B. Proteins
- C. Lipids
- D. Nucleic acids
- E. Water

39. The vitreous humor of eye is composed of _____

- A. Heparin
- B. Chondroitin sulfate
- C. Keratan sulfate
- D. Hyaluronic acid
- E. Dermatan sulfate

40. Which of the following is a glycerol based lipid?

- A. Phosphatidyl choline
- B. Ganglioside
- C. Sphingomyelin
- D. Vitamin D
- E. Prostaglandin

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42. All are biological functions of carbohydrate except:

- A. They serve as energy sources
- B. They used as source of carbon to synthesis non-essential amino acids
- C. They are involved in recognition and signaling for immune system
- D. They are used as structural materials, not protective components of the cells
- E. They conjugate with other biomolecule to give strength for tissues

43. The most frequent types of amino acid residues in collagen formation are:

- A. Glycine and proline
- B. Lysine and hydroxylysine
- C. Cysteine and Methionine
- D. Glycine and Lysine
- E. Alanine and Histidine

44. Which one of the following statements is correct regarding chaperones?

- A. They promote aggregation of non-native proteins
- B. They are synthesized only when they are induced by stimuli
- C. Usually recognize hydrophilic region of amino acid residues
- D. Associated with the degraded proteins during folding process
- E. They promote premature folding of proteins

45. During the formation of collagen protein, which of the following amino acids are commonly hydroxylated?

- A. Leucine and Isoleucine
- B. Cysteine and Methionine
- C. Proline and Lysine
- D. Arginine and Glycine
- E. Leucine and lysine

46. Among the following types of protein denaturing agents, which one is classified as physical denaturant?

- A. Extreme pH
- B. Ionizing radiations
- C. Heavy metals
- D. Organic solvents
- E. Detergents

47. The type of pentose sugar found in DNA molecule is:

D. 2-Deoxyribose

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48. The bonds involved in the formation of a single nucleotide of human RNA include:
- A. Glycosidic and phosphoester bonds
 - B. Phosphoester bond and hydrogen bonds
 - C. Electrostatic and hydrogen bonds
 - D. Phosphoester and disulfide bonds
 - E. Disulfide and ionic bonds
49. The nucleotide which is important for activation of monosaccharide units in polysaccharide synthesis is:
- A. Cyclic Guanosine monophosphate
 - B. Uridine Triphosphate
 - C. S-adenosyl methionine
 - D. Cyclic Adenosine Monophosphate
 - E. Phosphoribosylpyrophosphate
50. Which one is the smallest type of RNA among the following nucleic acids?
- A. Transfer RNA
 - B. Messenger RNA
 - C. Heterogeneous nuclear RNA
 - D. Ribosomal RNA
 - E. Both C and D
51. Which of the following groups of amino acids are mainly located in the inner core of globular proteins?
- A. Glutamate, Arginine, Lysine
 - B. Valine, Isoleucine, Glycine
 - C. Aspartate, Serine, Methionine
 - D. Histidine, Glutamine and Asparagine
 - E. Serine, Tyrosine and Glutamate
52. Identify the incorrect statement regarding the general characteristics of standard amino acids.

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53. The sulfur containing amino acid having non-polar side chain is
- A. Methionine
 - B. Alanine
 - C. Cysteine
 - D. Threonine
 - E. Selenocysteine
54. Which pairs of amino acid residues can undergo ionic interaction during the formation of protein tertiary structure?
- A. Lysine and Glutamate
 - B. Arginine and Valine
 - C. Glutamate and Aspartate
 - D. Lysine and Arginine
 - E. Proline and Valine
55. Which group of amino acids have similar side chain character based on the nature of polarity?
- A. Proline, Methionine, Leucine
 - B. Threonine, Valine, Proline
 - C. Glycine, Leucine, Asparagine
 - D. Phenylalanine, Glutamine, Alanine
 - E. Aspartate, Histidine, Alanine
56. Our body at adult stage can synthesize one of the following amino acids but not adequate during pregnancy.
- A. Methionine
 - B. Tryptophan
 - C. Leucine
 - D. Histidine
 - E. Asparagine
57. Which type of amino acid is important for the synthesis of the neurotransmitter, dopamine?
- A. Cysteine
 - B. Arginine
 - C. Isoleucine
 - D. Tyrosine
 - E. Glutamine
58. The pKa values of aspartic acid at α -carboxyl, side chains and α -amino are 2.0, 3.5 and 9.5 respectively. The approximate isoelectric point (PI) value of this amino acid is
- A. 2.75

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59. The primary structure of proteins can be described as
- A. the sequence of amino acid residues in a polypeptide
 - B. the type of hydrogen bonding and local folding pattern of a polypeptide
 - C. the type of interaction between different linear strands in a polypeptide
 - D. the association between different polypeptide units
 - E. covalent bonding between polysaccharides and polypeptides
60. Which one is not considered as advantage of quaternary structure formation in the native protein?
- A. Less number of amino acid are required for protein synthesis
 - B. Length of DNA sequence required to synthesize the oligomeric proteins
 - C. Formation of enzyme active sites using more than one protein subunit
 - D. Favors cooperative activity of protein subunits
 - E. Protection of hydrophobic residues for water soluble proteins